

Crop Recommendation Using Machine Learning Algorithms

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Abstract- The study presents an advisorial system on crops using Machine Learning (ML) for the extreme challenge which is the selection of suitable crops for cultivation depending on the varying climate and soil parameters at a regional level. This system links both the Indian Meteorological Department (IMD) and soil databases, and focuses on vital parameters such as soil pH, N(nitrogen), P(phosphorus), K(potassium), temperature, humidity, as well as precipitations. Data collection, data preprocessing by cleaning, normalization and imputation of missing values were taken into consideration, leading all to high-quality inputs Development of System. Several advanced machine algorithms including Random Forest, Support Vector Machines, decision trees, and Logistic Regression were performed to analyze parameters and suggest recommendations for crop predictions. Random forest achieves the highest accuracy, 99.3%. The system possesses an interface that runs on multiple languages to give access to farmers belonging to different geographies and dynamically responds to real-time weather data. Fertility advice, yield prediction, and risk assessment are utilized to drive farmer decision-making. The solution possesses a 15% enhanced prediction accuracy rate compared to existing systems to attempt to encourage sustainable agriculture practice by way of optimized crop recommendation and avoidance of excessive fertilizer application. The future direction of work would be scaling the system to multiple agro-climatic regions and integration of IoT-based sensors to gather real-time data to empower the farmers with more actionable intelligence.

Keywords—Machine Learning, Crop Prediction, Random Forest, Support Vector Machine, Decision Tree, Logistic Regression, Gradient Boosting, Precision Agriculture.

I. INTRODUCTION

Agriculture is one of the backbone sectors of the economy and contributes considerably to the GDP of the country. In India, in particular, agriculture supports the livelihood of millions of its people. Conventional farming over the years has resulted in repetition cropping and improper use of fertilizers due to which soil degradation and crop yield has significantly dropped. Therefore, it is imperative to utilize modern technologies like machine learning for working on holistic approaches for crop predictions and management of yield.

Machine learning, an artificial-intelligence sub-domain, enables machines to simulate intelligent behavior of human

beings and can perform some very complex tasks. The use of AI systems-that have implications similar to human reasoning-can aid agriculture activities. Machine learning begins with data collection-drawn in this case, from the financial transaction, personal information, or agricultural observations-and then it processes and uses it for training models. Generally, such models work more effectively with higher volumes of data.

With various types of soils in the Indian subcontinent, including alluvial soil-good for crops like cotton and rice, black soil-appropriate for crops like sugarcane and sunflower, red soil-for crops like corn and ragi, and laterite soil-best suited for pulses, tea, and coffee, crop recommendation needs to be region-specific. Hundreds of research studies have been conducted with the scope of optimizing agricultural planning. Crop recommendation using algorithms of machine learning is a need within such an exercise. The core focus of this study is to achieve a system in which farmers use data-driven decisions to choose the crops best suited for them according to soil conditions and weather. Several works apply machine learning models for crop recommendation and rainfall prediction. However, approaches proposed so far have some critical limitations.

[1] presented the Artificial Neural Network crop prediction model with parameters such as pH, nitrogen, temperature, and rainfall (document). It quite well can study the dynamics of crop yields among the environmental variables. However, its applicability is very narrow as it predicts only four crops such as cotton, sugarcane, wheat, and rice. Thus, having such a restricted focus point does not allow it to be applied broadly to agricultural practices, thus negating its application in other crops for making effective recommendations.

[2] developed a machine learning-based crop recommendation system, employing SVM and Decision Tree algorithms (crop-prediction-using-m.... Here, this model combined soil pH, temperature, and humidity to suggest crops. However, this method had its very own constraints. In this process, the use of SVM for temperature and humidity prediction is itself inadequate to get the accuracy of the rainfall predictions. The fact is that rainfall is such a really important aspect of determining whether crop production is viable or not. Their system did have a very simple interface, not so friendly,

especially for the illiterate farmer-cum-users of the system. Moreover, the system had no support to any language, making it unattractive to a large population of users in a multilingual country like India.

[3] have designed a crop prediction system by applying the machine learning algorithms such as Random Forest and Decision Trees. The motivation of the research was the improvement in the accuracy of crop yield predictions based on the variables like soil nutrients, temperature, and rainfall. Although the model was quite accurate, it had some disadvantages from the lack of diverseness in crops that could be predicted and the absence of direct integration of real-time datasets into the model. The same constraints limited its adaptability toward different climatic and regional conditions, which brought it to the need for far more dynamic and inclusive methods.

[4] suggested crop yield prediction models, including integration of big data frameworks like Hadoop along with machine learning algorithms. The paper was mainly on efficiency in managing big agricultural data sets to ensure accuracy in predictions for crop yields. Although this research provided good grounding to handle big data, the complexity and high cost associated with the infrastructure of setting up Hadoop made its practicality a challenge to achieve in small-scale agriculture implementation, hence reducing the practicality for widespread applications.

[5] developed a system that used soil data and rainfall forecasts based on machine learning algorithms to provide region-specific crop recommendations. In this work, it is brought into focus that rainfall and soil parameters decide the viability of crops. It was based on historical data and did not adapt to real-time changes in weather or environmental conditions, which made it less practical for dynamic agricultural needs.

[6] portrayed the role of machine learning in precision agriculture. His research used machine learning models with IoT devices to make real-time data collection systems and more accuracy in the predictions made. Because expensive infrastructure, it would not have been possible to reach many small farmers in the underdeveloped areas or be very scalable or even inclusive in his approach by improving the efficiency of agricultural sectors by avoiding unnecessary wastage of resources.

[7] worked on the concept of rainfall prediction using the algorithms of Support Vector Machine (SVM) and Relevance Vector Machine (RVM). This work states that rainfall was the very vital factor for the sustenance of crops, thereby indicating how the model predicts the rain with ease. Still, the research did not assimilate rainfall predictions with the soil data, thereby lessening the utility for comprehensive crop recommendation systems and opened scope for improvement in multi-parameter predictions.

[8] did an in-depth survey on the application of Decision Tree algorithms to the classification problem, including datasets from agriculture. Their study discussed the merits of the Decision Trees, which included interpretability and efficiency and thus suited crop recommendation systems. The survey is more inclined towards theoretical issues and never considered

the real implementation aspect or integrating the Decision Tree with real-world data sets from agriculture, thereby presenting an applied solution gap for farmers.

[9] have been proposed. In the said work, a friendly interface has been made available to the farmers who may obtain crop recommendations according to the soil and weather conditions. Although the proposed model successfully brought the facility of simple predictions within their reach, it lacked important facilities like rainfall prediction, etc., which made their output not very precise for applications on a large agricultural field.

[10] explored smart farming techniques by developing crop recommendation systems based on machine learning algorithms such as Random Forest. The research focus was given to the integration of weather and soil data for improved prediction. However, the system failed to address the accessibility issues as it is not accessible to the users who might need to receive assistance in multilingual, or an even more user-friendly interface.

[11] presented crop yield prediction done using machine learning algorithms, specifically regression models. The authors gave a proper understanding of how to do the agricultural output forecasting based on the historical data. However, the study only focused on the yield prediction and did not focus much on the crop recommendation part, nor it considered any regional specific factors like type of soil and weather condition, hence the applicability is very limited in practical farming scenarios.

[12] presented an IoT-based accuracy agriculture framework, that provided real-time data through various sensors integrated with the applications of machine learning algorithms that showed the feasibility of adding sensor-based inputs for greater accuracy in crop recommendation as well as agricultural planning; however, it still involved expensive sensor technology and associated IoT infrastructure, thereby preventing its access to the poorest smallholder farmers and marginalized regions in terms of further scalability.

[13] have proposed a hybrid approach using Decision Trees and Support Vector Machines for crop suitability predictions. The model is very much dependent on various multiple factors like soil pH, temperature, and rainfall so that the accuracy of crops selected improves. But, the system had difficulties while predicting dynamic weather patterns related to rainfall and also was unilingual or inaccessible to any user interface, thus less pragmatic for the mass users involved in farming.

[14] did a comparative study of machine learning algorithms for crop recommendation. The authors found that ensemble methods like Random Forest performed better than the simple models. The study was very helpful in deciding the choice of algorithm but did not address the issues of user interface design and practical implementation problems that are important for real-world applications.

[15] integrated the utilization of weather data along with machine learning models for an improved accuracy of crop choice. The research failed to emphasize soil-specific parameters, or the system's ease of access to the final users, which would have had a better overall efficiency for the system.

On the contrary, our approach utilizes a customized data set for the real-time IMD India Meteorological Department. We used this dataset in machine learning models, such as SVM and Decision Tree models, in order to overcome the lack of dynamism in datasets used previously and generalized. This system, besides offering increased precision towards the prediction of rainfall, will suggest more than four crops for the farmers. Moreover, our system will have a GUI with multiple language support, reaching out to farmers in more regions.

Our system, therefore, attempts to provide a more robust, inclusive tool for crop recommendation and rainfall prediction than these previous models, which address the shortcomings of these models.

II. METHODOLOGY

This proposed system includes weather in real-time data and machine learning algorithms, which may further arm this system to produce even more accurate crop recommendations than conventional systems. A customized dataset is built up using the weather data from IMD's website to make an output with a highly accurate prediction of precipitation, an almost essential feature in selecting crops. Currently, previous models were not so advanced or even more erroneous. This system will be able to offer easy and extensive range crop recommendation to the end-user.

A. Data Collection and Pre-processing

Data is fetched from the official website of IMD regarding weather parameters such as rainfall, temperature, and humidity, which become one of the vital factors governing crop yields. Local agricultural sources, which furnish the required data about soil, particularly regarding pH, nitrogen, phosphorus, and potassium, are used. This way, the model would make good use of historical data as well as real-time data to predict more correctly. The pre-processing step cleans the data, normalizes them, and cures them of missing values to assure uniform behavior when used in model training and testing.

B. Machine Learning Algorithms

The prediction algorithms of machine learning require highly accurate estimation using previously learned data. Predictive analytics historical information is the use of data, statistical methods, and machine learning approaches to forecast future results. It aims to transcend knowing what happened to giving the best possible solution and a prediction about what will happen next.

These include Logistic Regression, Decision Tree, Support Vector Machine, and Random Forest in this model.

1) Logistic Regression

Logistic Regression is one of the most widely applied statistical techniques for binary classification problems where the outcome variable is categorical with two possible outcomes: yes/no, 0/1, etc. Unlike linear regression, it is designed to model the probability of a binary response based on one or more predictor variables.

2) Decision Trees

Decision Trees (DT) are the classification and regression algorithms that fall under supervised learning. A tree is used for

the representation to solve the problem, with the leaf node representing a class label, and the interior node represents the attributes.

3) Support Vector Machine

A support vector machine is the very powerful supervised learning of an algorithm, used greatly in classification and regression, often effective in high dimensional space, and the model created by SVM is quite interesting because it creates the boundary of decision with maximal margins between different classes.

4) Random Forest

Random forest is an ensemble learning algorithm that produces millions of unique models to classify, or to solve any other issues of regression and more. It uses decision trees for training. The algorithm for random forest forms decision trees from multiple data samples, then it predicts data in each subset, then by voting, it provides a better solution to the system RF applies the bagging process for training the data to increase the accuracy of the result.

5) System Architecture

The system consists of several stages:

- **Data Input:** It allows real-time input data of soil pH, NPK values, and parameters of weather (temperature, humidity, predicted rainfall). The data can be inputted by the farmers manually or interfaced with sensors to collect the data automatically.
- **Crop Recommendation:** It uses the algorithm of Decision Tree that processes input data and gives the best crop recommendations for cultivation. Furthermore, it further guides on the amount of NPK required, estimated yields, and market prices anticipated for that respective crop.
- **Disease Detection :** Technology is applied in critical ways that include helping detect plant diseases so that the damaging of crops to a low extent is enabled and ultimately yields improve. Its use of ML technology analyzes plant images and information to assist in disease development and insights for efficient management.

C. User Interface and Accessibility

Major drawbacks in the system, such as in Nischitha's model, include a basic GUI with no support for multilingual. Our system will provide an easy user-friendly and interactive GUI which will be usable by all classes of users, and also will have multilingual support across the regions of India. This will help to ensure that the system will be technologically advanced, but accessible to farmers, who may not read and speak languages.

1) *Output and Recommendations* The final output of the system shall be:

- Recommended crop based on real-time data.
- Required NPK nutrients per hectare for the recommended crop.
- Estimated quantity of seed required per acre.
- Predicted yield and market price for the recommended crop.

The system will alert the users concerning possible risks, such as no rain for certain crops, to enable the users to make informed decisions.

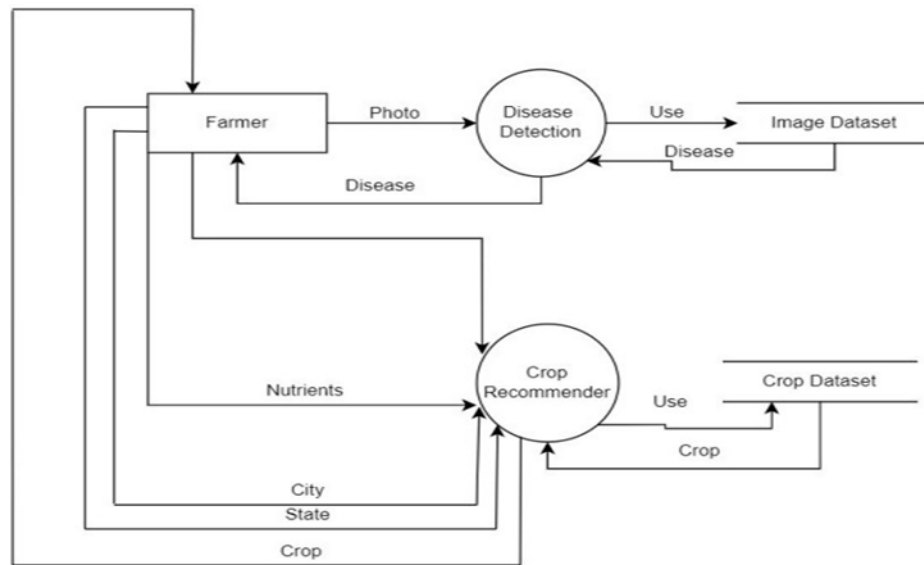


Fig. 1. Data Flow Diagram

III. RESULT

This research performs a comparative study on four machine learning techniques in classifying and predicting the suitable crop to grow in for different soil and climate condition such as Logistic Regression, Decision Tree, Support Vector Machine, and Random Forest. Soil samples are collected from various data sources, and this meteorological data is collected from the Indian Meteorological Department (IMD) based on conditions such as organization: minimum temperature, relative humidity, and rainfall. These are linked to soil chemical properties such as soil pH, nitrogen (N), phosphorous (P), and potassium (K) level. These climatic and nutrient soil factors affect plant growth, disease resistance, and overall performance in the agricultural scenario, as well as contribute to crop production and productivity determination.

To determine the predictive ability of these models, the data is divided into 80% training and 20% testing, thus having an even strategy towards model generalization. All the algorithms are trained on data and then tested based on the accuracy of classification to determine the model with the best crop recommendation. Logistic Regression achieves an accuracy of 97.6%, Decision Tree achieves 98.7%, Support Vector Machine (SVM) achieves 98.0%, and Random Forest achieves an accuracy of 99.3%. The improved performance by the Random Forest model is attributed to the ensemble nature of the model, which combines many decision trees to achieve robustness and prevent over fitting, thus resulting in higher reliability in crop prediction. The findings show that ensemble models, particularly Random Forest-based models, are effective in agriculture-related tasks since they are good at identifying subtle, nonlinear relationships in soil, weather, and yield. SVM, also known for its performance with high-dimensional data, is

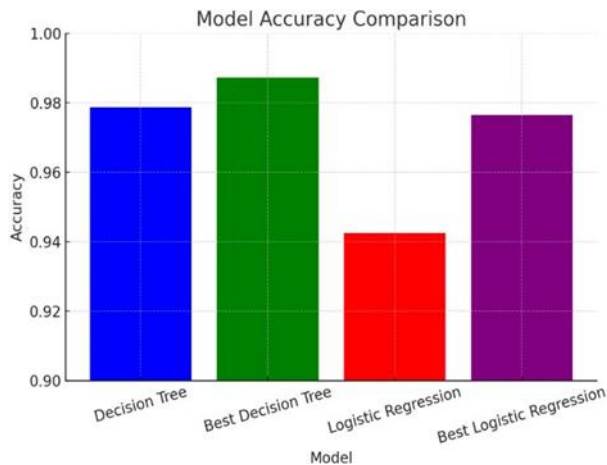
competitive but may be susceptible to the requirement of tuning of the hyper parameters and the kernel function for the attainment of even greater improvement. While interpret able and computationally friendly, Logistic Regression performs with lower accuracy compared to the others, i.e., the lower-dimensional linear approaches are perhaps not sufficient to capture all the underlying relationships in agriculture-related datasets.

Aside from model choice, the study also shows the significant contribution of preprocessing methods on data, such as feature scaling, outlier detection, and handling missing data, that significantly contribute to model performance. Because climate change and its impact on farm crops change dynamically, the incorporation of real- time weather data, satellite images, and remote sensing data can further enhance model precision. Future studies can include applying deep learning algorithms such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) to predict spatial and temporal interactions in agricultural data. Incorporating explainable AI (XAI) approaches also makes it possible to explain and interpret model outputs for farmers to be able to identify why certain crop recommendations are produced according to the circumstances.

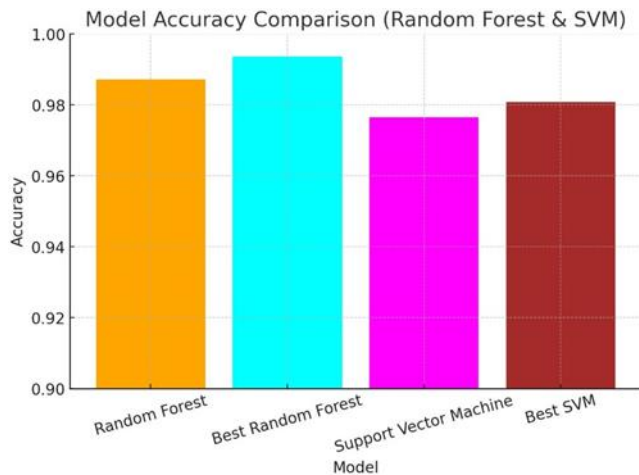
The research concludes that crop recommender systems based on machine learning can improve farm decision- making through data-driven suggestions, minimizing resource wastage, and maximizing yield potential. Integrating such models into mobile applications, decision support systems, or precision agriculture platforms can enhance farmers' usability and accessibility, which can lead to sustainable and technology-driven agriculture. The results for each algorithm were as follows:

TABLE 1. Algorithm and their accuracy table

Sr. No.	Algorithm Name	Accuracy
1	Logistic Regression	97.6
2	Decision Tree	98.7
3	SVM	98
4	Random Forest	99.3



(a)



(b)

Fig. 2. Model Accuracy

IV. CONCLUSION

This study results in the ability of machine learning algorithms, including Logistic Regression, Decision Tree, Support Vector Machine, and Random Forest, to predict the right crops to be grown at a specific soil and environmental data. Hence, in the comparisons presented by Table 6, Random Forest was determined to be accurate at a level that ensured sure recommendations for crops that would meet the district's agricultural conditions specifically.

The real-time weather information that is integrated from the India Meteorological Department (IMD) and also with the user-friendly interface having support for multiple languages, will help the farmer take better decisions on what to grow and in

which particular fertilizer to use. The output, encompassing required nutrients and estimated yields, delivers a more rounded form of decision making that thereby contributes to higher agricultural productivity at reduced costs.

This system is envisioned to be integrated with Internet of Things -based sensors for capturing real-time data. This would make further improvements in the accuracy of the crop recommendation and allow rectifications at appropriate times based on dynamic conditions surrounding the environment. A smarter agriculture in the future with IoT and machine learning would open up a more sustainable and resource-friendly ecosystem for farmers to change with the climatic conditions and maximize the usage of resources.

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